Total No. of Printed Pages: 1

SUBJECT CODE NO: -RNP-8068
FACULTY OF SCIENCE AND TECHNOLOGY
MDMT-2 Multidisciplinary Minor (MDM) (NEP) SEM-IV
Examination May/June 2026
Robotics Kinematics and Dynamics (Mechanical)

[Time:1:30 Hours] **[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B
1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from remaining Part B: Q.2 to Q.7.

Part A

Q.1 Solve any 5 of the following questions. **10**

a) What is an actuator? Give an example
b) What is a control system in robotics?
c) What is DOF of a manipulator?
d) Define link and joint.
e) What is continuous path motion?
f) What is robot programming?
g) Define velocity sensor.

Part B

Solve any four from the following.

Q.2 Describe various robot configurations (Cartesian, cylindrical, spherical, articulated). **05**

Q.3 Explain the working of proximity and velocity sensors. **05**

Q.4 Compare forward and inverse kinematics. **05**

Q.5 Explain with diagram Working of Magnetic Grippers. **05**

Q.6 Explain different types of Motion commands in VAL with example. **05**

Q.7 Explain Program with Delay & why it is used? **05**